

Enhancing Culinary Experience: Real-Time Interactive Cooking Assistance Utilizing Mixed Reality

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ABSTRACT

Advancements in 3D graphics and the integration of multimedia techniques in user assistance systems have emerged as a pervasive interest in recent research. Considering an environment and differences in individuals is significant for providing effective interactive directions. Several studies have proposed methods of presenting such guidance; however, the aforementioned factors are not taken into account. To build upon these studies, this paper presents an interactive cooking instruction application. The system is designed to guide users with user-adaptive features through the use of Meta Quest 3, a mixed reality (MR) based wearable device. To develop the system, the ingredients, cooking utensils, and cooking gestures are detected in real-time by a machine learning model to identify cooking processes. Through the proposed system, it is expected to provide new insight in the integration of culinary practices and MR, as well as in the approach of presenting interactive guidance of diverse fields via technical equipment.

Index terms: cooking assistance, user-oriented, mixed reality, object detection, hand gesture tracking and application

1 INTRODUCTION

Advancements in 3D graphics have led to significant developments in Virtual Reality (VR) and Augmented Reality (AR) technologies. These technologies create immersive experiences by blending real-world environments with computer-generated graphics. Alongside these advancements, there has also been progress in the associated hardware and software technologies. This research aims to develop a system utilizing the recently emerged Mixed Reality (MR) technology [1], a blend of VR and AR. This approach monitors the way users interact with cooking processes and offer guidance through user-friendly instructions. Traditional methods of cooking guidance, such as cookbooks, written recipes, or instructional videos, face significant limitations. They are inherently static and do not adapt to the user's environments or skill levels of individual users. This often forces users to adjust their cooking methods and kitchen setups to align with the instructions provided, which can be cumbersome and less effective. In order to address these challenges, this study takes inspiration from recent advancements in augmented reality-based cooking assistance systems. The system investigated by Isaias Majil identifies ingredients and suggests appropriate



Figure1: Real-time recognition of user's cooking environment detects necessary ingredients, utensils and guides the cooking process by detecting hand movements to complete the dish.

recipes without the need for external sensors or devices in the kitchen [2]. This system provides an interactive, step-by-step guide through an AR headset, making the cooking process more intuitive and personalized. Therefore, this research focuses on employing MR technology to overcome the inherent limitations of preexisting cooking recipe guidance. The proposed system is designed to make home cooking more accessible and engaging based on individual preferences and environments. This is done by recognizing the user's cooking actions and guiding them through virtual aids. This research will introduce the development of such system utilizing a MR technology-based wearable device that recognizes cooking processes in real-time and provides intuitive, user-oriented cooking guidance. The device that will be used is a Meta Quest 3 [3], developed by Meta, and a machine learning model trained with YOLOv8 will also be deployed for recognizing various ingredients and cooking utensils. Hand movements, during the action of cooking, are recognized using a hand shape recognition API provided by Meta [4]. To feature the guide for the cooking process, the system designed will be developed in Unity [5] and be displayed through the Passthrough of the meta device. This will allow the display to show directly in front of where the user is currently positioned in the environment. The novelty of this research lies in the usage of a wearable device with MR technology to guide cooking processes, without the need for any special markers or QR codes. The system will recognize both the ingredients and kitchen utensils, identify the current cooking action, and automatically progress to the next step in the cooking process. The contents of this paper are organized as follows. Section II presents a review of the relevant literature, emphasizing existing methods in the domain

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of cooking guidance and MR applications. Section III details the methodologies employed in the study, focusing on the detection of cooking ingredients, utensils, and hand gestures using machine learning models. Section IV discusses the design of the experiment, including the configuration of the application prototype and the metrics used to evaluate its performance. Finally, Section V summarizes the key findings of the study, highlighting the contributions and potential implications for future studies in this field.

2 RELATED WORK

Assorted studies have been designed to provide assistance for cooking such as smart kitchens based on the use of the Internet of Things (IoT) [6], a user-centric cooking support system [7], MR cooking support systems [8] and an augmented reality based interactive cooking guide [9].

Younes Fadil and others introduce an interactive application responsive to the user's movements in MR to assist with cooking [10]. The system selects a recipe by considering the detected ingredients and the user's preferences, recognizes cooking movements, and displays visual guidance. In the experimental setup, the authors set two cameras, a color camera, and an infrared camera, from a top-down view to detect objects. The specific ingredients and utensils are distinguished by utilizing paper markers which store information including the number of markers, name and color of the object, and the cooking action. In order to recognize the cooking actions, the positions and gravity centers of the user's hands are analyzed. When cutting is occurring, both hands move in the same direction at a constant distance. Once the distance between the hands exceeds a certain threshold, cutting is predicted to be finished. Mixing is assumed to be in progress when the hand movement traces a circular trajectory. When other actions such as "putting in" are done, the system detects markers, and subsequently, it compares the archived data to match the identified food with the corresponding cooking equipment. These methods are integrated into a parallel transition model to provide assistance. However, the research holds the following limitations: (1) The cameras restrict the cooking environments. (2) The markers are likely to become stained, resulting in lower recognition rates. (3) There are constraints to the possible positions for the markers. To address these constraints, this study seeks to identify specific cooking actions using a camera that moves according to the user's view without the need for physical markers. Apart from the aforementioned idea, each movement is detected by training the positional coordinates of specific hand parts.

A study investigated by Isaias Majil and others reduces the limitations of the previous research paper through the implementation of AR [11]. The study proposes a novel framework for providing cooking instructions by utilizing an AR wearable device, specifically, a headset. This system is fundamentally segmented into three primary phases: scanning of food ingredients, recipe recommendation, and delivery of a step-by-step cooking video tutorial. In the first phase, the built-in camera on the AR equipment detects the food automatically, and individual ingredients are then discerned through You Only Look Once (YOLO), a deep learning model. In the second stage, a list of recipes is presented according to the presence of ingredients. In the last phase, a video tutorial illustrating the performance of each cooking step of the chosen recipe is showcased through the utilization of AR glasses. Throughout the entire procedure, the designated hand gestures of the user dictate the execution of certain functions. Such functions are choosing a recipe, playing, and stopping the video, moving to the next step of the recipe, and closing the application, among other actions. This methodology integrates into diverse user environments without requiring

additional sensors or physical markers which effectively addresses the aforementioned constraints. Nevertheless, the study is subject to its own constraint. No other supplementary assistance is available beyond displaying of the video tutorial. To improve upon these restrictions, this paper proposes the utilization of tracking hands. This allows the provision of automatic instruction transfer and safety guidance on top of presenting visual tutorials.

Overall, this research introduces the following novelty: Recognizing each cooking action by leveraging the capabilities of Meta Quest 3, it is feasible to provide personalized cooking instructions tailored to the user's preferences in diverse environments.

3 METHODOLOGY

Reflecting on the studies reviewed in Section II, it becomes apparent that there is a limitation in recognizing objects or actions without the use of physical markers. Additionally, there has been no practical research on using AR devices to detect objects in the user's surrounding environment as it relates to automatically advancing through cooking stages. Therefore, this research has designed a system that recognizes surrounding objects and actions using a mobile, user-perspective camera, without the need for physical markers.

The proposed research is divided into four main sections: Object Detection, Hand Gesture Tracking, Networking, and App Development. Object detection focuses on detecting ingredients and cooking utensils. Hand gesture tracking is used for detecting hand movements to identify cooking actions during cooking.

In the section on networking, this research uses real-time passthrough video streamed by Meta Quest 3 headset to a server to gather input images for object detection and hand tracking. Passthrough is a feature that allows the user to step outside the user's view in VR to see a real-time view of the user's surroundings [12]. Lastly, app development involves developing this system into an application and designing a user interface for end-users. When the user wears a Meta Quest 3 headset, the real-time passthrough footage seen through the device is transmitted to the server via a wireless network connection. The server, using a pre-trained custom model, analyses the footage sent from Meta Quest 3 in real-time. In this process, the server identifies the ingredients, cooking utensils, hand gestures, and cooking motions in each frame of the footage. The server also uses the captured footage, processing it with the YOLOv8 model, and sends the results in JSON format back to the Meta Quest 3 via a UDP socket. Furthermore, this system is developed as an application that can be executed on the Meta device, which provides various recipes and suggests necessary ingredients for such. It recognizes ingredients and tools placed in the user's kitchen environment, by determining their presence or absence. Then, the system divides the cooking process into stages, detecting necessary cooking actions at each stage. As the next action starts, the system automatically transitions to the next stage, thus guiding the user to complete the cooking process.

To validate this idea, Aglio e Olio Pasta is selected as the menu item for proof of concept. Aglio e Olio Pasta requires simple and few ingredients while demanding only a small variety of cooking gestures and skills, which makes the recipe easy to replicate.

3.1 Object Detection

To generate the datasets, this research collects images of ingredients and cooking utensils. Through machine learning training, a custom model that recognizes the ingredients and cooking tools placed in the user's kitchen environment during the

cooking process is generated, so there is no need for markers or sensors. The object detection technology recognizes each food component and tool through the Passthrough of a user wearing a Meta Quest 3 headset. Table 1 depicts ingredients and cooking utensils that need to be recognized during the preparation of Aglio e Olio Pasta.

Table 1. Labels and classes for ingredients and utensils

Label Num.	Utensils	Label Num.	Ingredients
0	Cutting Board	5	Boiled Spaghetti
1	Knife	6	Garlic
2	Pot	7	Raw Spaghetti
3	Plate	8	Salt
4	Tongs	9	Pepper
		10	Olive Oil

Two categories are considered for selecting the image data: ‘Universal images’ and ‘user-taken images.’ Universal images are not considered within the cooking environment defined for this study. Primarily, these are images that can be obtained through online search platforms. For the collection of universal images in this study, Roboflow Universe [13] is utilized, and data with annotations provided in this platform are used. User-taken images are considered more in-line with the specific cooking environment and purposes of this study. For these images, videos of actual Aglio e Olio Pasta preparation are recorded from the perspective of a user wearing a Meta Quest 3. These videos are then converted into frame-by-frame photos to be used as image data. They are manually annotated with the use of the labellmg [14] program, and the order of labels and class names are also specified manually.

These two datasets of image data are created separately with the same condition which includes label order and class names. Each dataset is trained through machine learning and the recognition rates and performance in an actual cooking environment are compared.

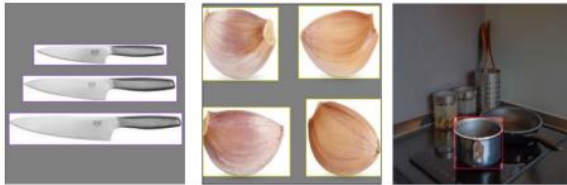


Figure 2: Example of image data for universal dataset

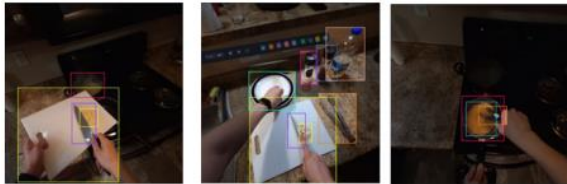


Figure 3: Example of image data for user-taken dataset

For the machine learning training, YOLOv8x is used for the object detection model. YOLOv8x stands out as a highly capable model in image processing among the currently available open-source options. There are five kinds of YOLOv8 models based on performance and size: YOLOv8n, YOLOv8s, YOLOv8m,

YOLOv8l, and YOLOv8x [15]. Among these, YOLOv8x is chosen to maximize performance and recognition rates.

3.2 Hand Gesture Tracking

This study aims to recognize the hand geometry shapes performed by users while cooking Aglio e Olio Pasta. Among the recognized hand shapes, those involved in the cooking processes are detected and specific movement paths of these hand shapes are tracked. This tracking helps identify which cooking action, such as stirring or chopping, the user is performing and helps the system interpret the progress of cooking. In addition, this system provides the users with the correct and safe hand positions during the chopping process. It identifies whether the shape of the hand that is holding the ingredients is safe or dangerous. Such ability is to assess the possibility of the finger being cut or injured by the knife. Table 2 presents a table summarizing the hand shapes and cooking actions that need to be detected for preparing Aglio e Olio.

Table 2. Labels, classes for hand shapes and gestures

Label Num.	Utensils	Label Num.	Ingredients
0	Open	0	Stop
1	Close	1	Stirring (Clockwise)
2	Grabbing Knife	2	Stirring (Counterclockwise)
3	Grabbing Tongs	3	Move
4	Finger Guard	4	Chopping
5	Dangerous		

To track hand gestures, TensorFlow API within the MediaPipe framework is utilized. The MediaPipe [16] package, developed by Google, provides high-precision hand tracking and gesture recognition. This enables accurate detection of hand positions, movements, and shapes in real-time. This is done with hand landmarks it provides after processing. The MediaPipe Hand Landmarker detects key points of hands in an image i.e., landmarks, allows the identification of hand positions and the application of visual effects [17]. Also, it achieves fast and efficient processing for real-time applications, making it well-suited for interactive and live environments.

For training the dataset, the Keras framework API [18] from TensorFlow [19] is deployed. Keras provides a user-friendly programming interface, enabling the rapid and easy construction of complex deep learning models. This approach allows for the easy creation of neural networks with multiple layers and the training of models using data.

The open-source code developed by Kinivi [20] is used for training in recognizing hand shapes and cooking motions. The code employs MediaPipe and the OpenCV [21] library to identify and track the key points of finger joints. In this process, a Multi-Layer Perceptron (MLP) from the neural network is utilized. The footage captured through the camera is converted into an RGB matrix frame by frame using OpenCV. The captured frames are cropped and used for the data of the hand detection model. The data is then transferred to a feed-forward neural network by TensorFlow Keras (see Figure 4). The neural network that is trained on the labeled dataset is responsible for learning a map between the input and output by adjusting its internal parameters such as weights and biases. In this stage, the position coordinates of designated key points (input) are classified to differentiate specific hand shapes (output).



Figure 4: Key points in order, example of hand shape recognition and patterns of movement path

To recognize cooking gestures, the system displays the patterns of the movement path for specific key points when certain hand shapes form, such as grabbing a knife or tongs. The patterns of the movement path appear with decreasing size as subsequent histories are displayed, which facilitates the distinction between the beginning and end of the movement path. Frames showing these path patterns are captured to generate a dataset. The system then classifies these patterns to identify the specific cooking gesture being done. To facilitate the cooking process of Aglio e Olio, this study has defined the necessary hand shapes and cooking gestures. Six hand shapes are required for the process of cooking Aglio e Olio. The two primary shapes include the hand holding a knife for cutting ingredients, and the hand grabbing tongs for stirring boiling pasta. For safety features during chopping, two hand shapes are defined: One that safely holds the ingredients to avoid cutting fingers and another that shows a more dangerous shape with an elevated risk of knife injuries. To distinguish unclear or irrelevant hand shapes, which could reduce the recognition rate of the earlier stated hand shapes, the list of recognized shapes also includes a clenched fist and a hand with all fingers fully extended.

Five gestures have been defined as necessary for cooking. The gesture of the knife-held hand moving up and down, while the other hand supports the ingredients, is defined as ‘Chopping.’ When the hand holding the tongs stirs the pasta or ingredients clockwise, it is defined as ‘Stirring 1’, and counterclockwise as ‘Stirring 2.’ To contrast movements that could interfere with the recognition of the three mentioned cooking gestures, irregular and random movements are categorized as ‘Move.’ And the absence of detected movement, indicating a static state, is represented as ‘Stop.’

3.3 Connecting MR Device with Server

This paper uses the Meta Quest Developer Hub [22] application, provided by Meta, to program the MR device. Through this application, the entire interface and passthrough screen displayed on the Meta Quest 3 is cast to a server computer. The mirrored cast screen is captured frame by frame using Python code, and object detection is performed using YOLOv8. As the system detects the objects, the label names and coordinate information are drawn on each detected object. The processed data is then normalized and converted to a JSON string format. The converted data is then sent to the application running on Meta Quest 3 via a UDP socket. The application on Meta Quest 3 receives this JSON file and converts it into a C# class. This data is then used to display the results of both ingredient detection and hand gesture tracking on the screen of the application. The UDP protocol is selected because it is suitable for real-time application due to no hand shaking process.

4 IMPLEMENTATION

Through object detection, the necessary ingredients are recognized, and hand gesture tracking is used to detect hand shapes along with cooking gestures, as discussed in Section 3. Derived from the object detection data and tracking data, this study developed an application that operates on a Meta Quest 3 headset.

The application provides cooking assistance to the user by processing a real-time video feed sent from Meta Quest 3 to a server. The server runs the custom model for a multifaceted recognition scheme and sends the detection data back to the Meta Quest 3

4.1 Object Detection

To obtain custom-trained models for recognizing ingredients and cooking tools, this study generates two models: Model A, which uses a .pt file trained with the universal dataset discussed in Section 3, and Model B, which uses a .pt file trained with a user-taken image dataset. These custom model files are tested with the same video to compare the results of object detection. The test video features a user wearing a Meta Quest 3 headset while cooking Aglio e Olio, captured from a passthrough perspective. To avoid overlap between the training dataset and the testing video, the video chosen for the test is intentionally selected to have a different kitchen layout than the one used for annotating the user-taken dataset. The training parameters for both datasets are set identically with epochs of 50, batch size of 16, and image size of 640.

In Figure 5 and Figure 6, the comparison of the same frames of the video shows significant differences between Model A and Model B. These differences are particularly evident in how each model recognizes specified objects.

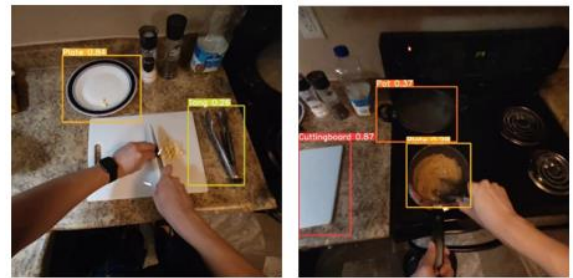


Figure 5: Recognition results for Model A



Figure 6: Recognition results for Model B

Model B, acquired by a user-taken dataset, is observed to detect significantly more elements for object recognition compared to Model A trained by the universal dataset. Model A barely identifies objects present in the overall test video, resulting in the lack of bounding boxes. Furthermore, the objects recognized by Model B matches with the actual objects, while those recognized by Model A often do not correspond to the real objects, leading to low recognition rates. Model A consists of images with objects clearly visible and easily identifiable, whereas Model B includes objects that appear in actual cooking processes. Taking this into account, Model B demonstrates relatively higher performance and accuracy. This is attributed to its ability to reflect diverse variables and scenarios found in kitchen environments, such as recognizing

ingredients on a plate or detecting pasta being cooked in a pot for the established environment.

4.2 Hand Gesture Tracking

Regarding the hand shapes, it is found that the recognition accuracy for grabbing a knife and holding tongs is significantly low due to similarities in the detected keypoint patterns. It is found that the main distinguishing feature lies in keypoint 6, which is the first joint of the index finger. When grabbing a knife that is characterized by its long and thin spine, keypoint 6 generally points vertically upwards. In contrast, when gripping tongs, fingers including keypoint 6 tend to wrap around, pointing horizontally outwards due to the wide grip. Taking this into account, collecting accurate data occurred when the index finger of the hand holding the knife is kept completely stretched instead of being bent, as shown in Figure 7. This is done to enhance the distinctiveness and recognition accuracy of this hand shape.

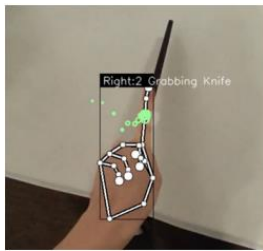


Figure 7: Recognition results for Model A

After training the collected dataset, the custom model is tested on videos of preparing Aglio e Olio from the user's perspective wearing the Meta Quest 3. The results indicate that the modified hand shape for holding a knife appears to have significantly improved recognition rates compared to the original shape. It is also clearly distinguishable from the hand shape for holding tongs. The safety feature recognition during chopping determines that the hand shape is dangerous when the tip of the thumb (keypoint 4) is closer to the knife than the second joints of the index, middle, and ring fingers (keypoints 7, 11, 5). The hand position is considered safe when keypoint 4 is further from the knife than keypoints 7, 11 and 5. This is demonstrated in Figure 8.

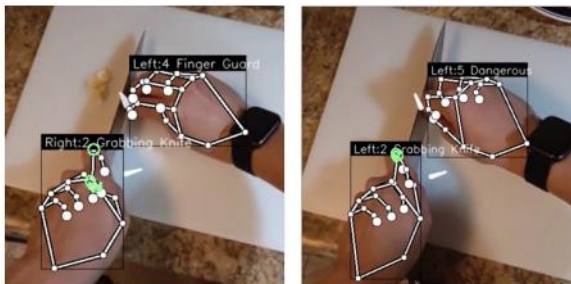


Figure 8: Example of safety feature while chopping

The recognition of cooking gestures shown in Figure 9 is also worth noting. When the hand shape for holding a knife is recognized, a path pattern from keypoint 6 tracks the movement of the hand. If the hand holding the knife moves up and down, creating a short vertical pattern in the path, it is recognized as 'Chopping.' Likewise, when the hand shape for holding tongs is recognized, a

path pattern from keypoint 5 is activated. If this pattern appears in a consistent circular shape around a central point, the system recognizes this gesture as 'Stirring.' It is observed that irregular circular shapes in the movement path or inconsistencies in the centre point and the radius of the circle lead to a result of inadequate recognition of stirring.



Figure 9: Example of chopping, stirring gesture recognition

4.3 Networking

The system utilizes the UDP protocol to send real-time data to the Meta Quest 3 by using a video stream transmitted to a server computer for YOLO object detection and hand tracking. When Meta Quest 3 sends a video stream to the server, the server captures the projected window and interprets this as the input for both YOLO object detection and the hand tracking process. The results from YOLO and hand tracking are stored in a dictionary type variable. Upon receiving an 'OK' message from the application, these results are converted into a JSON string and transmitted back to the application via a UDP socket. When the application receives this JSON string, it extracts the YOLO and hand tracking results, which stores them in a public variable. This data is then accessed and utilized by other scripts as needed. This data is then accessed and utilized by other scripts as needed, e.g. "IngredientManager" script for detecting ingredients and "InstructionManager" script for detecting cooking actions. Additionally, the application sends back an 'OK' message to continue receiving data for subsequent frames.

4.4 App Development

The application is developed on a Windows 11 PC and uses Unity 2022.3.12f1 for the 3D application engine. The system incorporates features like Passthrough and hand tracking from the Oculus Integration asset [23] to use the functionalities of the Meta Quest 3. Upon executing the application, the system initiates a UDP socket communication to start receiving data from the server. The data arrives in the form of a JSON string. This is then processed into a usable C# class format using the JSON function from Unity's JSON utility [24]. Users are able to view the external environment through the passthrough layer which is shown with a UI panel. Utilizing Meta's Hand Tracking API [25], users are able to control a pointer with hand movements and make selections through a pinch gesture. The Hand Tracking API enables users to interact with an application, especially for manually navigating through cooking steps.

The application begins with the user clicking the start button, leading to a menu screen. The application proceeds to the ingredient and tool preparation phase. During this stage, lists of required ingredients and utensils are displayed in a scroll view. Using the YOLO result data received via UDP communication, the application identifies visible objects in the user's field of view. When necessary, ingredients and tools are detected, the text labels of corresponding items in the lists are highlighted by changing their

text colour. Once all the materials are prepared, a button to proceed to the next stage becomes activated, enabling the start of cooking instruction as shown in Figure 10.

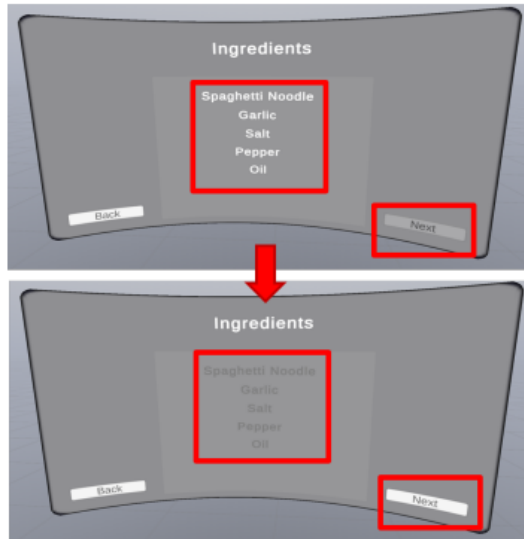


Figure 10: Ingredients panel

To address situations where YOLO does not successfully identify some objects, the system allows users to manually highlight the items by clicking on their text labels. In the instruction phase, the application presents users with instructions and walk through videos as demonstrated in Figure 11. The application automatically progresses to the next stage after a certain duration once the user has initiated the corresponding action. This progression is based on hand gesture data received from the server, which identifies the user's actions. The system also allows the users to manually navigate to the next or previous stages using arrow buttons, enabling them to check earlier steps. The program ends after completing the cooking process by following all the given instructions.

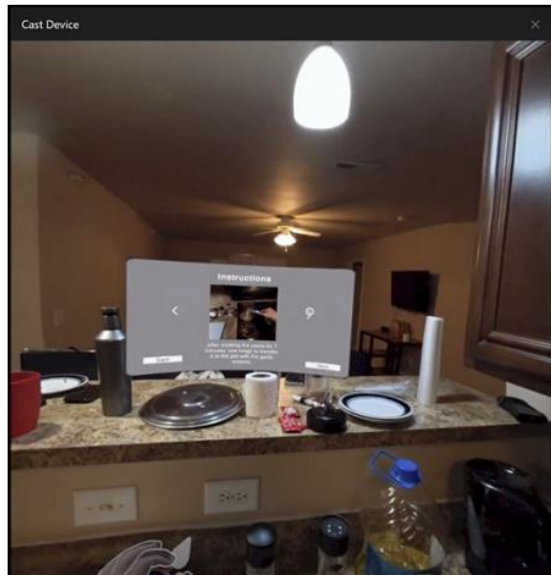


Figure 11: Instruction panel

4.5 Discussion

This paper develops a system to assist users in cooking utilizing mixed reality (MR) technology. The proposed system recognizes kitchen utensils and cooking ingredients via object detection without additional markers. It also recognizes users' cooking motions through hand gesture tracking. Based on the recognized contextual information, the system provides corresponding guidance and instructions to users via the MR headset in an interactive manner. The results validate the potential of employing MR devices for interactive cooking assistance. Future work includes improving the recognition accuracy of ingredients using object detection algorithms, incorporating more varieties of ingredients and hand motions to enhance interactivity and context awareness. Thorough studies on these will contribute towards building more intelligent and practical MR cooking assistant systems that can provide users with more accurate culinary information and instructions.

5 CONCLUSION

This study explores the use of a MR environment to assist users with cooking and instruction. The collected datasets were trained to detect the ingredients, utensils, and cooking actions. Based on the results, automatic recognition of necessary components in food preparing steps were feasible. Following this, a feature that determines the safe hand position was explored and deemed viable. Compared with other approaches, this research introduces a novelty by utilizing a system in a dynamic environment that effectively guides and instructs users. It also provides this interactive experience without the use of physical markers. Future research into this topic and the application of such technology is expected to facilitate more advanced metrics in the cooking context while also exploring safety features that could be used to protect users such as fire detection. Culinary practices will see development as a result of these studies into integration of technology and the art of cooking.

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